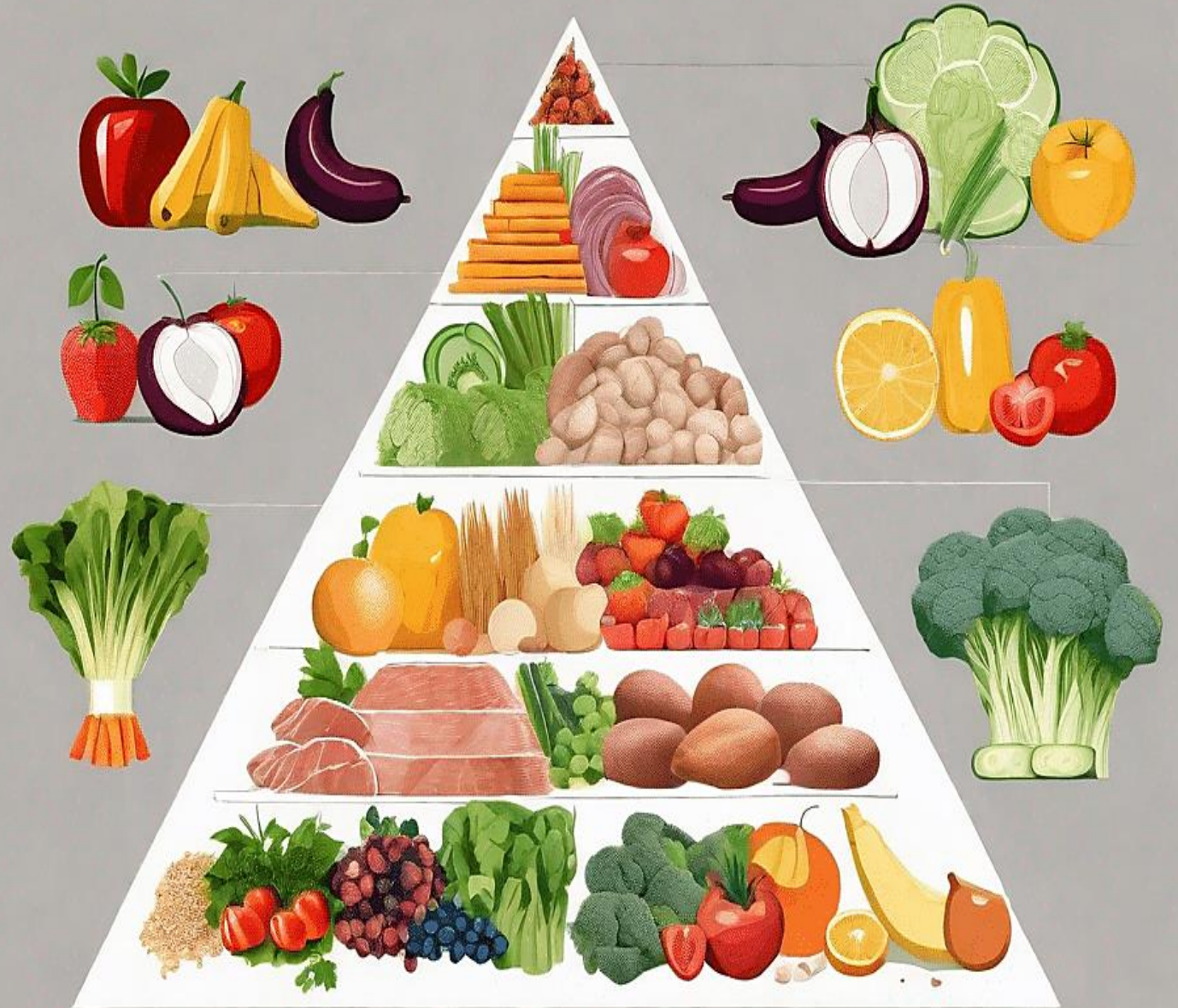


Nutrition



Learning Objectives:

- Understand the Concept of Nutrition
- Identify the Role of Macronutrients
- Examine the Functions of Carbohydrates
- Explore the Role of Proteins
- Understand the Function of Fats
- Explore the Role of Micronutrients
- Nutritional assessment
- Role of nurse to enhance balanced nutrition



Nutrition

process by which the body **takes in**, **digests**, **absorbs**, and **utilizes food and nutrients**.

Purposes of healthy nutrition

- To **maintain health**
- To **support growth**
- To **provide energy**
- To **enable proper bodily function**.

Components of nutrition

- **Macronutrients:** Nutrients that the body requires in large amounts to provide normal bodily functions.

There are three primary types of macronutrients:

1. carbohydrates

2. proteins

3. fats

- **Micronutrients:** Nutrients required by the body in small amounts for proper growth, development, and overall health support various physiological processes but do not provide energy directly. It includes:

1. Vitamins

2. minerals

Macronutrients

Macronutrients	Functions or importance	Types		sources
Carbohydrates	❖ Main source of energy.	Simple	Complex	❖ Bread ❖ Pasta ❖ Rice ❖ Potatoes ❖ Fruits ❖ Vegetables ❖ legumes
		➤ Found in foods like fruits, milk, and sweeteners like sugar.	➤ Found in foods like whole grains, vegetables, and legumes.	
		➤ Quickly digested	➤ Digested more slowly	
		➤ Provide a rapid source of energy.	➤ Provide sustained energy.	

Macronutrients

Macronutrients	Functions or importance	Types		Sources
Proteins	▪ Source of energy, especially when carbohydrates are not available ▪ Essential for building and repairing tissues ▪ Producing enzymes and hormones ▪ Supporting immune function	Proteins are made up of amino acids.		• Meat • Fish • Eggs • dairy products • beans • Nuts • Seeds • tofu
		Essential amino acids	Non-essential amino acids	
		Body can't make them and must obtained from food.	Formed in body	

Macronutrients

Macronutrients	Functions or importance	Types			Sources
Fats	➤ Provide a concentrated source of energy ➤ Support cell function ➤ Help the body absorb fat-soluble vitamins (A, D, E, K). ➤ They are also involved in hormone production ➤ Protecting vital organs.	Saturated fats	Unsaturated Fats	Trans Fats	▪ Butter ▪ Oils ▪ Avocados ▪ nuts ▪ Seeds ▪ fatty fish (like salmon) ▪ cheese
		Found in animal products as butter, cheese, and some plant oils	Found in plant-based oils, nuts, seeds, and fatty fish	Artificial fats found in processed foods	
		Provide energy,	Help reduce the risk of heart disease		
		Excessive intake can contribute to health issues like heart disease.	Considered healthier fats	Increase the risk of heart disease.	

Micronutrients

- **Vitamins:** organic compounds that help regulate vital processes in the body .

Classification of vitamins

Fat-Soluble Vitamins	Water-Soluble Vitamins
Stored in the body's fat tissues and liver	Not stored in the body
Absorbed with the help of dietary fat	dissolve in water
Released as needed.	Require regular dietary supply
Stored for long periods	Excess amounts are excreted through urine, so they need to be replenished regularly.
Examples: A, D, E, K	Examples: B Vitamins (B1, B2, B3, B5, B6, B7, B9, B12) and Vitamin C

Fat soluble vitamins

Vitamins	Functions	Sources
A (Retinol)	➤ Essential for vision ➤ Immune system function ➤ Skin health.	▪ Carrots, sweet potatoes, kale, liver, eggs, dairy.
D (Calciferol)	➤ Helps the body absorb calcium and phosphorus. ➤ Promoting bone health ➤ support immune function.	▪ Sunlight, fortified dairy products, fatty fish (salmon, tuna), egg yolks.
E (Tocopherol)	➤ Acts as an antioxidant protecting cells from damage. ➤ Supporting immune function.	▪ Nuts, seeds, vegetable oils, broccoli.
K (Phylloquinone)	➤ Vital for blood clotting ➤ Bone health.	Leafy green vegetables (kale, spinach), broccoli, fish, meat.

Water-Soluble Vitamins

Vitamins	Functions	Sources
Vitamin C (Ascorbic Acid)	➤ Supports the immune system. ➤ Helps with the absorption of iron. ➤ Promotes wound healing. ➤ Acts as an antioxidant.	▪ Citrus fruits (oranges, lemons), strawberries, bell peppers, broccoli.
B Vitamins B1 (Thiamin) B2 (Riboflavin) B3 (Niacin) B5 (Pantothenic Acid) B6 (Pyridoxine) B7 (Biotin) B9 (Folic Acid) B12 (Cobalamin)	These vitamins play a crucial role in: ➤ Energy production ➤ Brain function ➤ Formation of red blood cells.	▪ Whole grains, meat, eggs, dairy, legumes, and leafy greens.

Minerals

- Minerals are **inorganic nutrients** that are essential for the body's structure and **functions**.

Classification of minerals

Major Minerals (Macro minerals)	Trace Minerals (Microminerals)
▪ Required by the body in larger amounts.	• Required in smaller amounts but are still essential for health.
➤ Calcium ➤ Phosphorus ➤ Magnesium ➤ Sodium ➤ Potassium ➤ Chloride ➤ Sulfur	➤ Iron ➤ Zinc ➤ Iodine ➤ Copper ➤ Selenium ➤ Manganese ➤ Fluoride

Major Minerals (Macro minerals)

Minerals	Calcium	Phosphorus	Magnesium	Sodium
Function	Essential for: ➤ Strong bones and teeth ➤ Nerve transmission ➤ Muscle function.	➤ Works with calcium to form strong bones and teeth. ➤ Involved in energy production.	➤ Supports muscle and nerve function. ➤ Helps with energy production. ➤ Involved in the synthesis of protein and DNA.	Regulates: ➤ fluid balance ➤ nerve function ➤ muscle contraction. (Note: Too much sodium can contribute to high blood pressure.)
Sources	▪ Dairy products (milk, cheese), leafy greens, fortified plant milks, tofu.	▪ Meat, fish, eggs, nuts, legumes.	▪ Nuts, seeds, whole grains, leafy green vegetables.	▪ Salt, processed foods, canned soups, and meats.

Major Minerals (Macro minerals)

Minerals	Potassium	Chloride	Sulfur
Function	Maintains: ➤ proper muscle and nerve function. ➤ fluid balance. ➤ blood pressure regulation.	➤ Works with sodium to maintain fluid balance. ➤ plays a role in digestion.	Important for protein synthesis and enzyme function.
Sources	▪ Bananas, potatoes, oranges, spinach, tomatoes.	▪ Table salt, sea salt, tomatoes.	▪ Garlic, onions, vegetables, eggs.

Trace Minerals (Microminerals)

Minerals	Iron	Zinc	Iodine	Copper
Functions	➤ Essential for the production of hemoglobin, which carries oxygen in the blood.	➤ Supports immune function, wound healing, and DNA synthesis	➤ Essential for thyroid hormone production, which regulates metabolism.	➤ Involved in iron metabolism, immune function, and the formation of red blood cells.
Sources	▪ Red meat, poultry, fish, legumes, spinach.	▪ Meat, shellfish, legumes, seeds, nuts, dairy.	▪ Iodized salt, seafood, dairy products.	▪ Shellfish, nuts, seeds, whole grains.

Trace Minerals (Microminerals)

Minerals	Selenium	Manganese	Fluoride
Functions	➤ Acts as an antioxidant and supports thyroid function.	Supports: ➤ bone health. ➤ wound healing. ➤ metabolism of carbohydrates, proteins, and fats.	➤ Helps to prevent dental cavities and supports bone health.
Sources	▪ Brazil nuts, seafood, meats, grains.	▪ Whole grains, nuts, legumes, leafy vegetables.	▪ Fluoridated water, tea, fish.

A doctor in a white lab coat with a blue stethoscope around their neck is seated at a desk, writing on a clipboard with a black pen. In the foreground, a variety of fresh fruits and vegetables, including red and yellow bell peppers, a green cucumber, a sliced kiwi, red apples, and an orange, are arranged on a reflective surface. A yellow rectangular box with a black border is positioned in the upper right quadrant of the image, containing the text "NUTRITIONAL ASSESSMENT" in bold, black, uppercase letters.

NUTRITIONAL ASSESSMENT

Nutritional Assessment

- **Dietary History**
- **Dietary Assessment Tools**
- **Anthropometric Measurements**
- **Biochemical Indicators**
- **Clinical Observation**

1. Dietary History

Dietary history involves gathering comprehensive information about :

- Patient's usual food intake
- Dietary habits
- Behaviors.

This may include details about meals, snacks, portion sizes, and any special dietary practices or restrictions.

Aspects of a dietary history include:

- **Meal Timing and Frequency:** What times does the patient typically eat? How many meals and snacks do they consume daily?
- **Food Preferences and Restrictions:** Does the patient follow any specific diets (e.g., vegetarian, gluten-free, low-sodium)? Are there any food allergies or intolerances?
- **Portion Sizes:** How much food does the patient typically eat at each meal?
- **Food Preparation:** Does the patient prepare meals at home, or do they rely on pre-packaged or restaurant foods?

Aspects of a dietary history include:

- **Alcohol Consumption:** Is alcohol consumed regularly? How much, and how often?
- **Hydration:** Does the patient consume adequate water or fluids? What types of drinks (e.g., soda, tea, coffee) are most common?
- **Eating Patterns:** Does the patient skip meals, have irregular eating habits, or engage in emotional or binge eating?

Common Dietary Assessment Tools

These tools help obtain information from patients about their food habits and nutritional status.

24-Hour Dietary Recall

- **Description:** The patient is asked to recall everything they have eaten and drunk in the past 24 hours, including portion sizes, cooking methods, and any snacks.

Common Dietary Assessment Tools

- **Food Diary (Food Record)**

- **Description:** The patient records all the foods they consume over a specified period (usually 3-7 days). This can be done by the patient or a trained professional.

- **Food Frequency Questionnaire (FFQ)**

- **Description:** The patient is asked to report how often they consume a variety of foods over a specific period (e.g., daily, weekly, monthly). This tool often includes a list of common foods and their portion sizes.

Common Dietary Assessment Tools

- **Diet History Interview**

- **Description:** This tool involves a structured interview where the patient provides detailed information about their food intake, preferences, and dietary patterns. A healthcare professional asks open-ended questions to gather qualitative data on food choices and habits.

- **Nutrient Analysis Software**

Description: This tool involves inputting the patient's food intake into a computer program or online platform that analyzes the nutritional content of foods, including macronutrients, micronutrients, and caloric intake.

Anthropometric Measurements

Anthropometric measurements **provide important data on a patient's nutritional status by assessing physical body measurements**. Common methods include:

- **Body Mass Index (BMI):** A measure of weight in relation to height. It is used to assess whether a patient is underweight, overweight, or at a healthy weight.

Formula: $BMI = \text{weight (kg)} / \text{height (m}^2\text{)}$

Categories:

- Underweight (<18.5),
- Normal weight (18.5-24.9),
- Overweight (25-29.9),
- Obesity (30+)

Anthropometric Measurements

- **Waist Circumference:** Used to **assess abdominal fat distribution**, which is linked to increased risk of cardiovascular diseases and metabolic disorders.
- **Skinfold Thickness Measurement:** Measures subcutaneous fat using calipers **to estimate total body fat percentage**.
- **Mid-Upper Arm Circumference (MUAC):** A measure of nutritional status, often **used to assess muscle mass and fat reserves**.



- **Waist-to-Hip Ratio (WHR):** This ratio compares the circumference of the waist to the hips and is used **to assess the distribution of body fat**.
 - Formula: $WHR = \frac{\text{Waist circumference}}{\text{Hip circumference}}$
 - A higher ratio can indicate an increased risk of cardiovascular disease.

Waist-to-Hip Ratio (WHR)



Waist circumference (cm)



Hip circumference (cm)

Waist-to-Hip Ratio (WHR) Norms				
Gender	Excellent	Good	Average	At Risk
Males	<0.85	0.85–0.89	0.90–0.95	≥0.95
Females	<0.75	0.75–0.79	0.80–0.86	≥0.86

Biochemical Indicators

- **Laboratory Tests:** Blood and urine tests can be used to assess nutrient deficiencies or imbalances. Examples include:
 - **Serum Albumin:** A protein in the blood that helps assess overall nutrition and liver function.
 - **Hemoglobin and Hematocrit Levels:** Used to detect anemia or iron deficiency.
 - **Vitamin and Mineral Levels:** Specific blood tests can measure the levels of important micronutrients, such as Vitamin D, calcium, or iron.

Clinical Observation

Healthcare providers often assess visible signs of malnutrition or nutrient deficiencies through **physical examination. Symptoms such as:**

- **Paleness, brittle nails, or hair loss** (possible iron deficiency).
- **Wounds that heal slowly** (could be a sign of protein deficiency or zinc deficiency).
- **Swollen or bleeding gums** (indicating Vitamin C deficiency or scurvy).
- **Dry, flaky skin** (may suggest essential fatty acid or Vitamin A deficiency).

Role of the Nurse in Promoting Healthy Nutrition

- Nurses play a vital role in promoting healthy nutrition and supporting patients in achieving and maintaining optimal health through proper dietary practices. They are often in a unique position to influence patient health outcomes because they interact with patients regularly, provide education, and coordinate care across various settings (hospital, community, home health, etc.).

Here are some roles nurses perform in promoting healthy nutrition:

Nutritional Assessment

Nurses are responsible for assessing the nutritional status of patients, which involves:

- **Collecting Dietary History:** Nurses gather information on patients' eating habits, food preferences, allergies, special dietary needs, and any current issues (e.g., difficulty swallowing, poor appetite).

Nutritional Assessment

- **Monitoring Anthropometric Measurements:** Regular measurements such as weight, height, BMI, and waist circumference are important for assessing nutritional status and identifying any risks for malnutrition or obesity.
- **Screening for Nutritional Deficiencies:** Nurses identify early signs of nutritional deficiencies (e.g., vitamin or mineral deficiencies) by observing clinical signs (such as skin changes, hair loss, or fatigue) and reviewing laboratory results (such as low hemoglobin or albumin levels).
- **Using Screening Tools:** Nurses use standardized tools, such as the Malnutrition Screening Tool (MST), the Mini Nutritional Assessment (MNA), or other dietary questionnaires, to detect malnutrition risk.

Patient Education

Nurses provide education on the importance of good nutrition for maintaining health and managing chronic conditions. **Some key aspects of nutritional education include:**

- **Promoting Healthy Eating Habits:** Nurses teach patients about balanced diets, proper portion sizes, and the importance of a variety of foods. They may educate about macronutrients (proteins, carbohydrates, fats) and micronutrients (vitamins and minerals), emphasizing how these contribute to overall well-being.
- **Hydration Education:** Nurses educate patients about the importance of staying hydrated and the role of fluids in maintaining proper bodily functions.

Patient Education

- **Managing Special Diets:** For patients with specific health conditions such as diabetes, hypertension, or celiac disease, nurses explain the need for specialized diets (e.g., low-sodium, low-carb, gluten-free) and help patients understand how to manage these dietary changes in daily life.
- **Dietary Modification for Weight Management:** Nurses offer guidance on healthy weight loss, healthy eating habits, and physical activity to patients struggling with obesity or overweight. They emphasize portion control, mindful eating, and reducing processed foods.

Advocacy and Collaboration

Nurses often work as part of an interdisciplinary healthcare team, collaborating with dietitians, physicians, social workers, and other professionals. They advocate for patients by:

- **Referring to Dietitians:** Nurses may refer patients to registered dietitians for more in-depth nutritional assessments and personalized diet plans. Dietitians can help manage specific nutritional needs, such as tube feeding, food allergies, or complex metabolic conditions.

Advocacy and Collaboration

- **Working with Physicians:** Nurses collaborate with physicians to monitor patients with conditions like diabetes, hypertension, cardiovascular disease, and cancer. They help physicians ensure that nutrition interventions align with medical treatment plans.
- **Patient Advocacy:** Nurses advocate for access to proper nutrition resources, such as meal assistance programs, nutritional supplements, or financial support for patients unable to afford healthy food.

Encouraging Healthy Lifestyle Choices

Nurses play an essential role in helping patients make healthy lifestyle changes that support overall well-being, including:

- **Promoting Physical Activity:** Nurses educate patients about the role of exercise in managing weight, preventing chronic diseases, and improving overall health. They may recommend activities that fit a patient's lifestyle and physical limitations.

Encouraging Healthy Lifestyle Choices

- . **Encouraging Preventive Care:** Nurses encourage regular screenings and preventive health measures, such as monitoring cholesterol, blood pressure, and blood sugar levels. They highlight the link between healthy eating and the prevention of chronic conditions like heart disease, stroke, and diabetes.
- . **Behavioral and Emotional Support:** Nurses provide counseling and emotional support for patients facing barriers to healthy eating, such as food insecurity, lack of time, emotional eating, or stress. They may refer patients to counseling or community resources if needed.

Monitoring and Documentation

Nurses are responsible for regularly monitoring a patient's nutritional status throughout treatment and documenting their progress:

- . **Tracking Nutritional Intake:** Nurses document patients' food and fluid intake, including any challenges such as nausea, vomiting, loss of appetite, or difficulty chewing/swallowing. This information is used to adjust care plans.

Monitoring and Documentation

- **Documenting Nutritional Interventions:** Nurses document any dietary recommendations or interventions provided to the patient, including specific dietary plans or changes in food orders (e.g., pureed diet, low-sodium, diabetic-friendly meals).
- **Evaluating Outcomes:** Nurses evaluate the effectiveness of nutrition interventions by monitoring weight gain/loss, improvement in laboratory values (e.g., blood glucose or serum albumin), and patients' reported well-being.

Managing Nutrition in Specific Populations

Nurses play an important role in managing the nutritional needs of specific patient groups, including:

- **Infants and Children:** Nurses assist with the introduction of solid foods, breastfeeding, and the introduction of nutritious meals. They help parents understand appropriate portion sizes, vitamin requirements, and food choices.
- **Pregnant and Lactating Women:** Nurses provide education on increased caloric intake, balanced meals, and nutrients important during pregnancy, such as folic acid, iron, and calcium. They also help manage issues like morning sickness and food aversions.

Managing Nutrition in Specific Populations

- **Elderly Populations:** Nurses assist elderly patients who may be at risk for malnutrition due to loss of appetite, chewing/swallowing difficulties, or medication side effects. They educate caregivers and families on meal planning, nutrient-dense foods, and the importance of maintaining a healthy weight in older age.
- **Patients with Chronic Conditions:** Nurses assist patients with chronic conditions like diabetes, cardiovascular disease, or gastrointestinal disorders to manage their diets and prevent complications. They ensure that patients understand how their nutrition affects their condition and support adherence to specialized diets (e.g., low-sugar, low-sodium, or gluten-free diets).

Nutritional Support in Clinical Settings

In clinical settings, nurses often play an essential role in the management of patients who require nutritional support, including:

- **Enteral Nutrition (Tube Feeding):** Nurses help administer and monitor enteral nutrition for patients unable to eat by mouth. They ensure that tube feedings are properly delivered, that patients are tolerating the feedings, and that the tube is functioning well.

Nutritional Support in Clinical Settings

- **Parenteral Nutrition (IV Nutrition):** In severe cases, patients may require intravenous (IV) nutrition, and nurses manage the preparation and administration of parenteral nutrition. This includes monitoring blood glucose levels, fluid balance, and infection risk.
- **Monitoring Nutritional Supplements:** Nurses are responsible for administering nutritional supplements (e.g., protein shakes, vitamins) as ordered and evaluating their effectiveness in improving nutritional intake.

